

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

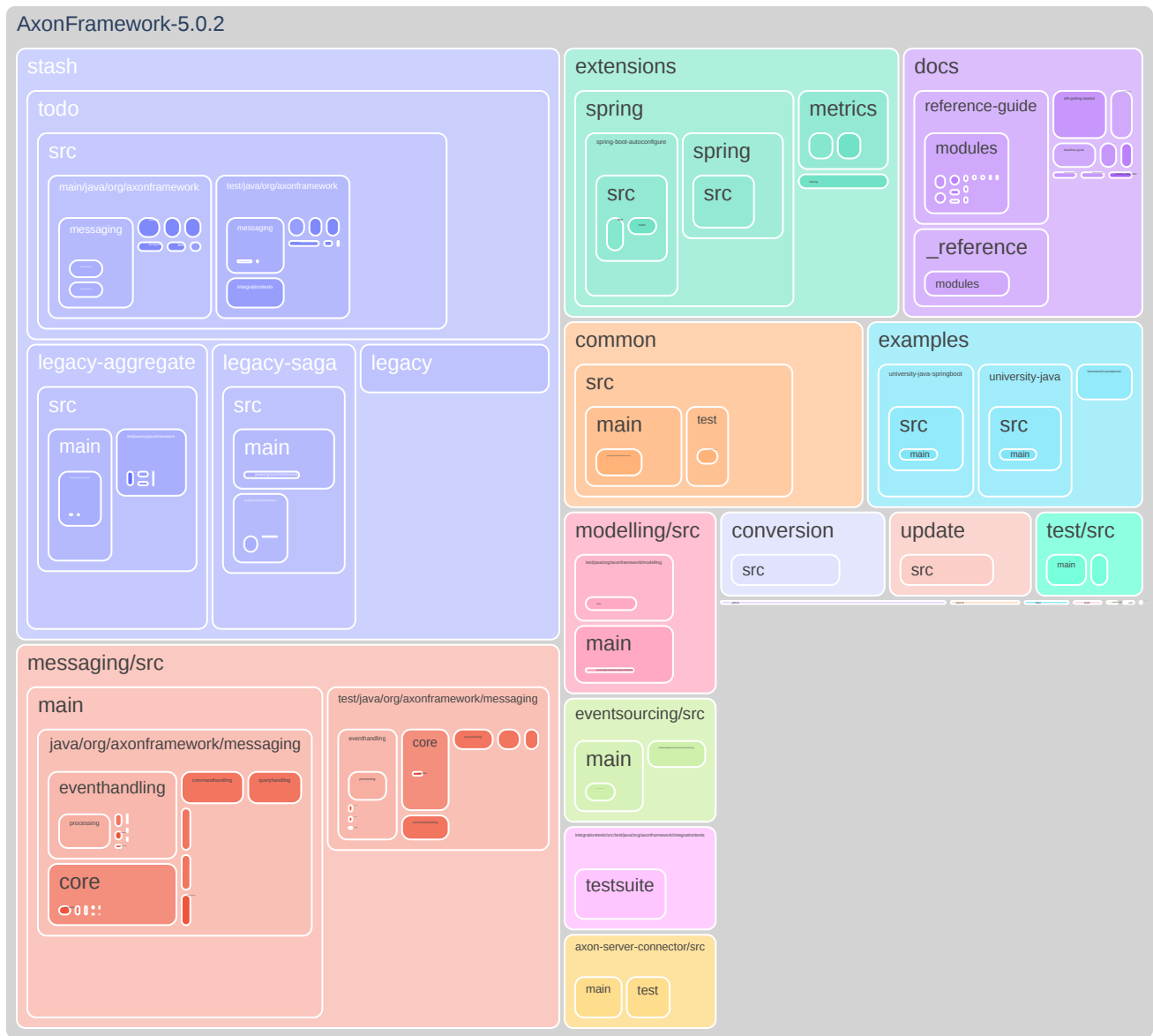
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	725.000000	725.000000	725.000000	725.000000	725.000000	725.000000
mean	22.416552	11.197241	171.793103	60.124138	130.911724	101.110345
std	112.370074	12.384873	363.609595	146.368022	185.012096	151.977459
min	1.000000	2.000000	3.000000	8.000000	11.000000	8.000000
25%	2.000000	4.000000	22.000000	14.000000	71.000000	18.000000
50%	4.000000	7.000000	44.000000	15.000000	82.000000	71.000000
75%	11.000000	13.000000	159.000000	15.000000	98.000000	82.000000
max	2640.000000	115.000000	6018.000000	545.000000	2753.000000	599.000000

	directoryPath	directoryParentPath	directoryName
0	AxonFramework-5.0.2/.idea/copyright	AxonFramework-5.0.2/.idea	copyright
1	AxonFramework-5.0.2/.idea/inspectionProfiles	AxonFramework-5.0.2/.idea	inspectionProfile
2	AxonFramework-5.0.2/axon-framework-bom	AxonFramework-5.0.2	axon-framework-bom
3	AxonFramework-5.0.2/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-5.0.2/axon-server-connector/src/main	resources/META-INF/service
4	AxonFramework-5.0.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/command	AxonFramework-5.0.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector	command
5	AxonFramework-5.0.2/axon-server-connector/src/test/resources	AxonFramework-5.0.2/axon-server-connector/src/test	resource
6	AxonFramework-5.0.2/build/coverage-report	AxonFramework-5.0.2/build	coverage-report
7	AxonFramework-5.0.2/build/parent	AxonFramework-5.0.2/build	parent
8	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/digest	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common	digest
9	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/io	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common	io
10	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common/tx	AxonFramework-5.0.2/common/src/main/java/org/axonframework/common	tx
11	AxonFramework-5.0.2/common/src/main/resources/META-INF/services	AxonFramework-5.0.2/common/src/main	resources/META-INF/service
12	AxonFramework-5.0.2/common/src/test/java/org/axonframework/common/lifecycle	AxonFramework-5.0.2/common/src/test/java/org/axonframework/common	lifecycle
13	AxonFramework-5.0.2/common/src/test/resources/META-INF/services	AxonFramework-5.0.2/common/src/test	resources/META-INF/service
14	AxonFramework-5.0.2/conversion/src/main/resources/META-INF/services	AxonFramework-5.0.2/conversion/src/main	resources/META-INF/service
15	AxonFramework-5.0.2/docs/_playbook/localLinks	AxonFramework-5.0.2/docs/_playbook	localLink
16	AxonFramework-5.0.2/docs/_reference/modules/ROOT/attachments	AxonFramework-5.0.2/docs/_reference/modules/ROOT	attachment
17	AxonFramework-5.0.2/docs/_reference/modules/ROOT/partials	AxonFramework-5.0.2/docs/_reference/modules/ROOT	partial
18	AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector/pages	AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector	page
19	AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector/partials	AxonFramework-5.0.2/docs/_reference/modules/axon-server-connector	partial
20	AxonFramework-5.0.2/docs/_reference/modules/configuration/pages	AxonFramework-5.0.2/docs/_reference/modules/configuration	page
21	AxonFramework-5.0.2/docs/_reference/modules/configuration/partials	AxonFramework-5.0.2/docs/_reference/modules/configuration	partial
22	AxonFramework-5.0.2/docs/_reference/modules/disruptor/pages	AxonFramework-5.0.2/docs/_reference/modules/disruptor	page
23	AxonFramework-5.0.2/docs/_reference/modules/disruptor/partials	AxonFramework-5.0.2/docs/_reference/modules/disruptor	partial
24	AxonFramework-5.0.2/docs/_reference/modules/eventsourcing/pages	AxonFramework-5.0.2/docs/_reference/modules/eventsourcing	page
25	AxonFramework-5.0.2/docs/_reference/modules/eventsourcing/partials	AxonFramework-5.0.2/docs/_reference/modules/eventsourcing	partial
26	AxonFramework-5.0.2/docs/_reference/modules/legacy/pages	AxonFramework-5.0.2/docs/_reference/modules/legacy	page
27	AxonFramework-5.0.2/docs/_reference/modules/legacy/partials	AxonFramework-5.0.2/docs/_reference/modules/legacy	partial
28	AxonFramework-5.0.2/docs/_reference/modules/metrics/pages	AxonFramework-5.0.2/docs/_reference/modules/metrics	page
29	AxonFramework-5.0.2/docs/_reference/modules/metrics/partials	AxonFramework-5.0.2/docs/_reference/modules/metrics	partial

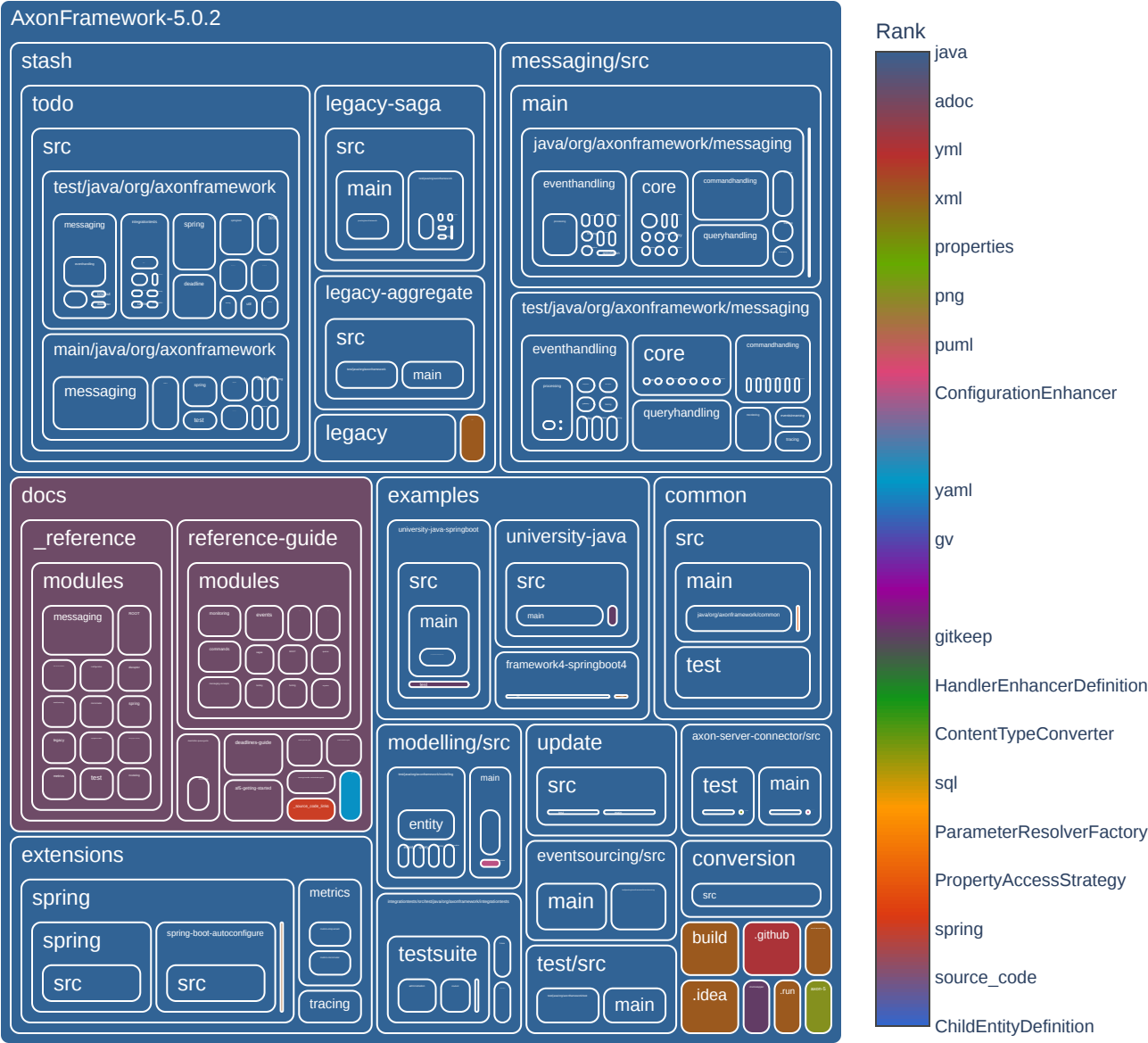
Number of files per directory

Directories and their file count



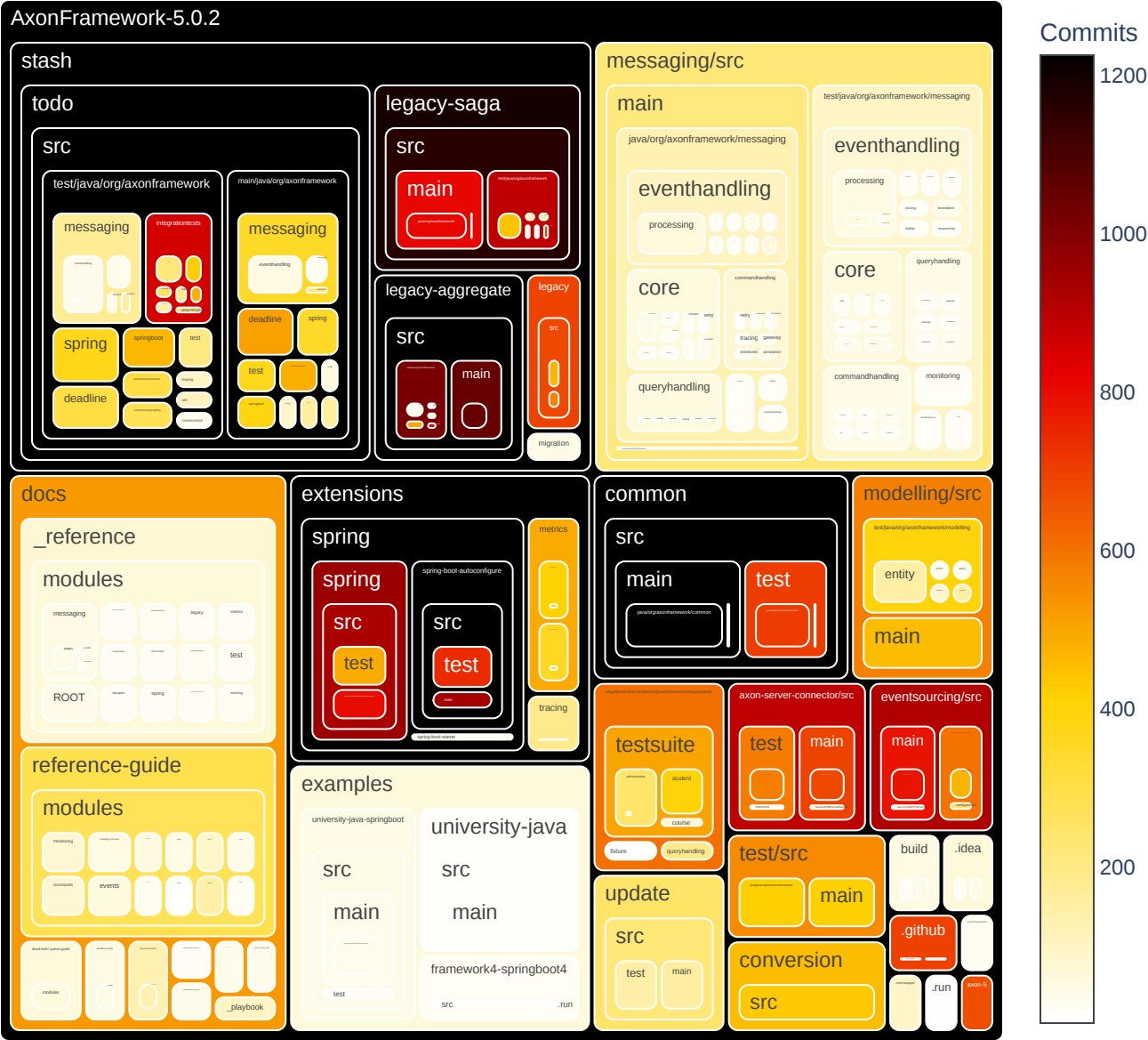
Most frequent file extension per directory

Most frequent file extension per directory



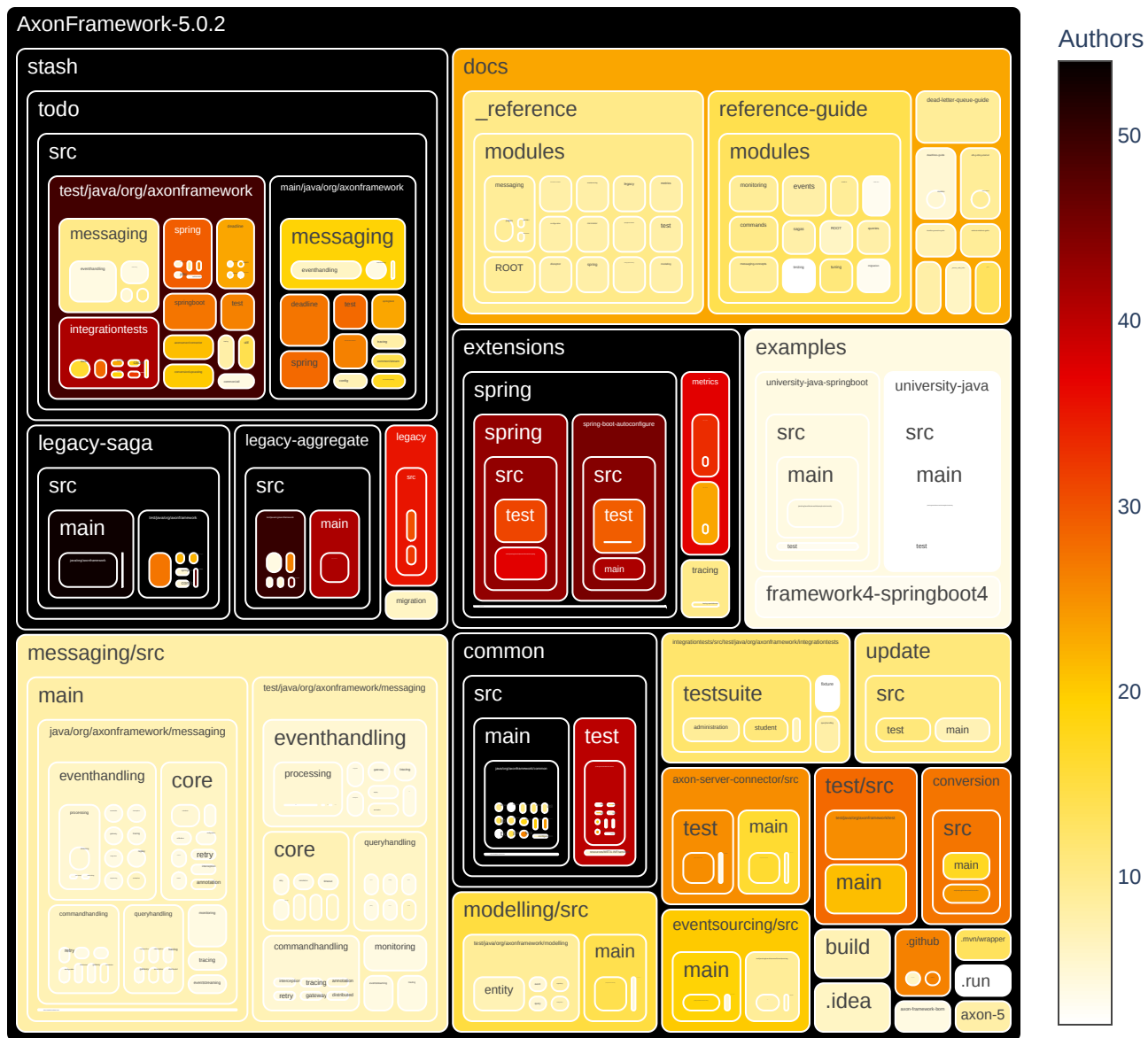
Number of commits per directory

Number of git commits



Number of distinct authors per directory

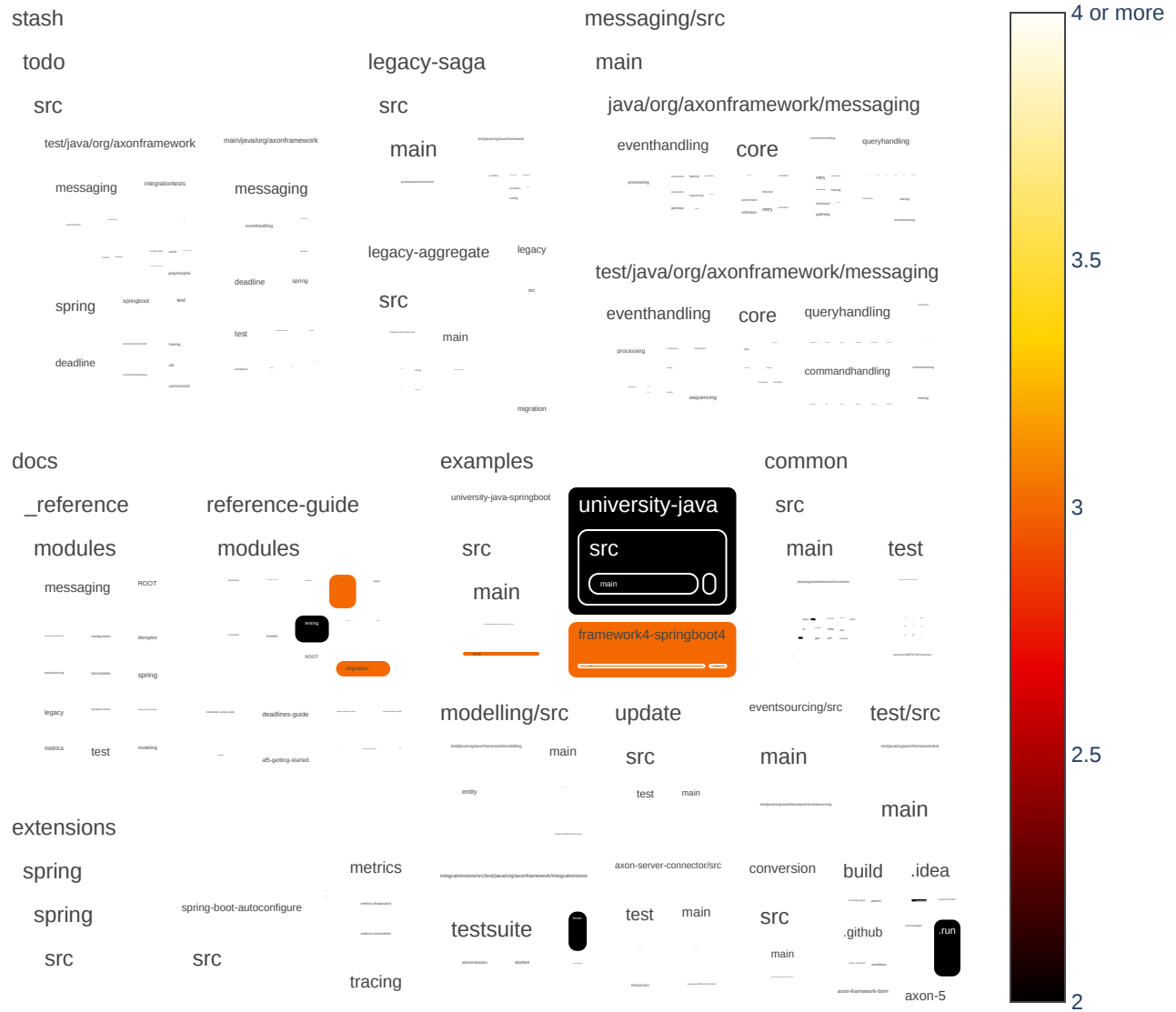
Number of distinct commit authors



Directories with very few different authors

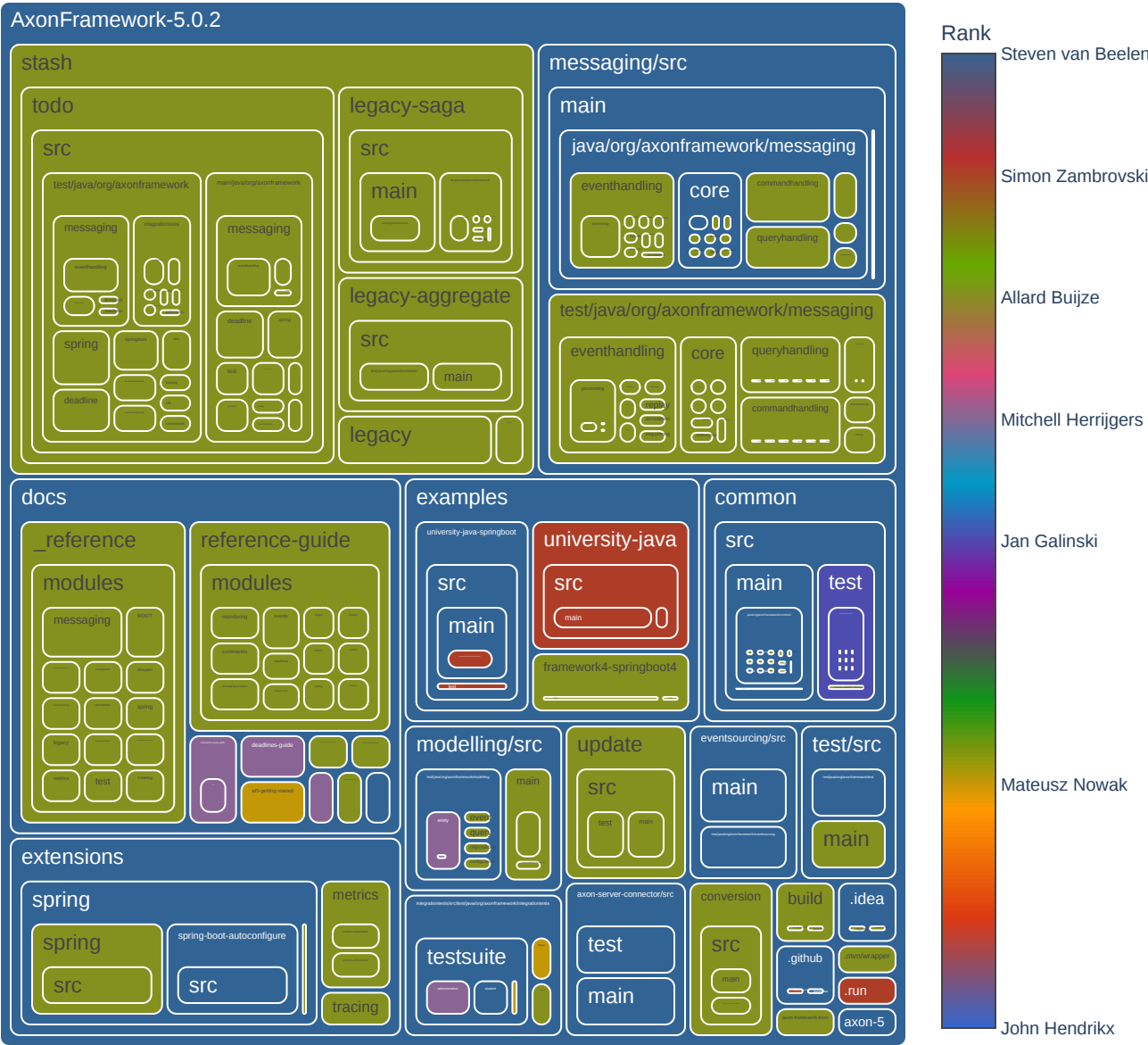
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-5.0.2



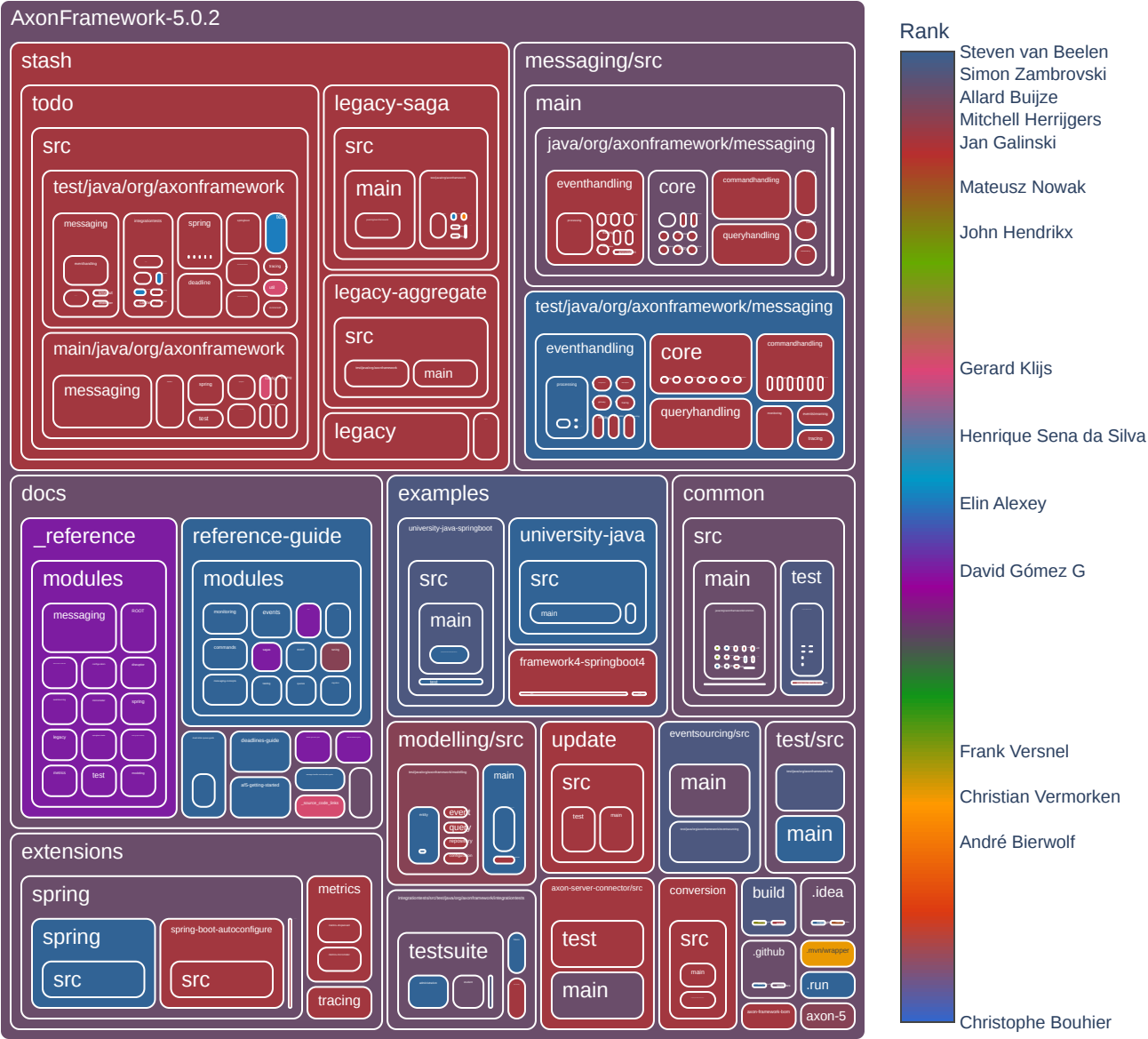
Main author per directory

Main authors with highest number of commits



Second author per directory

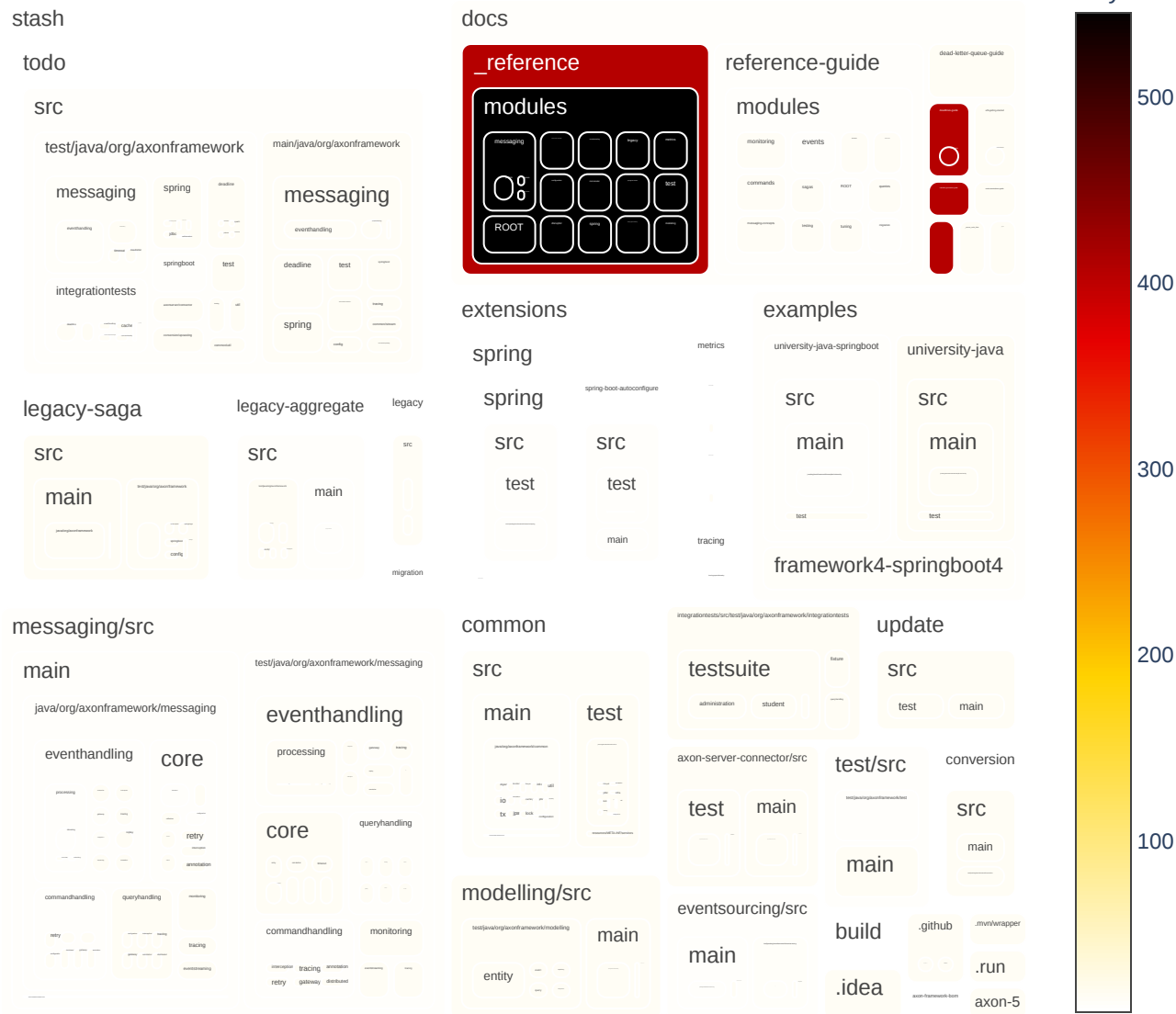
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

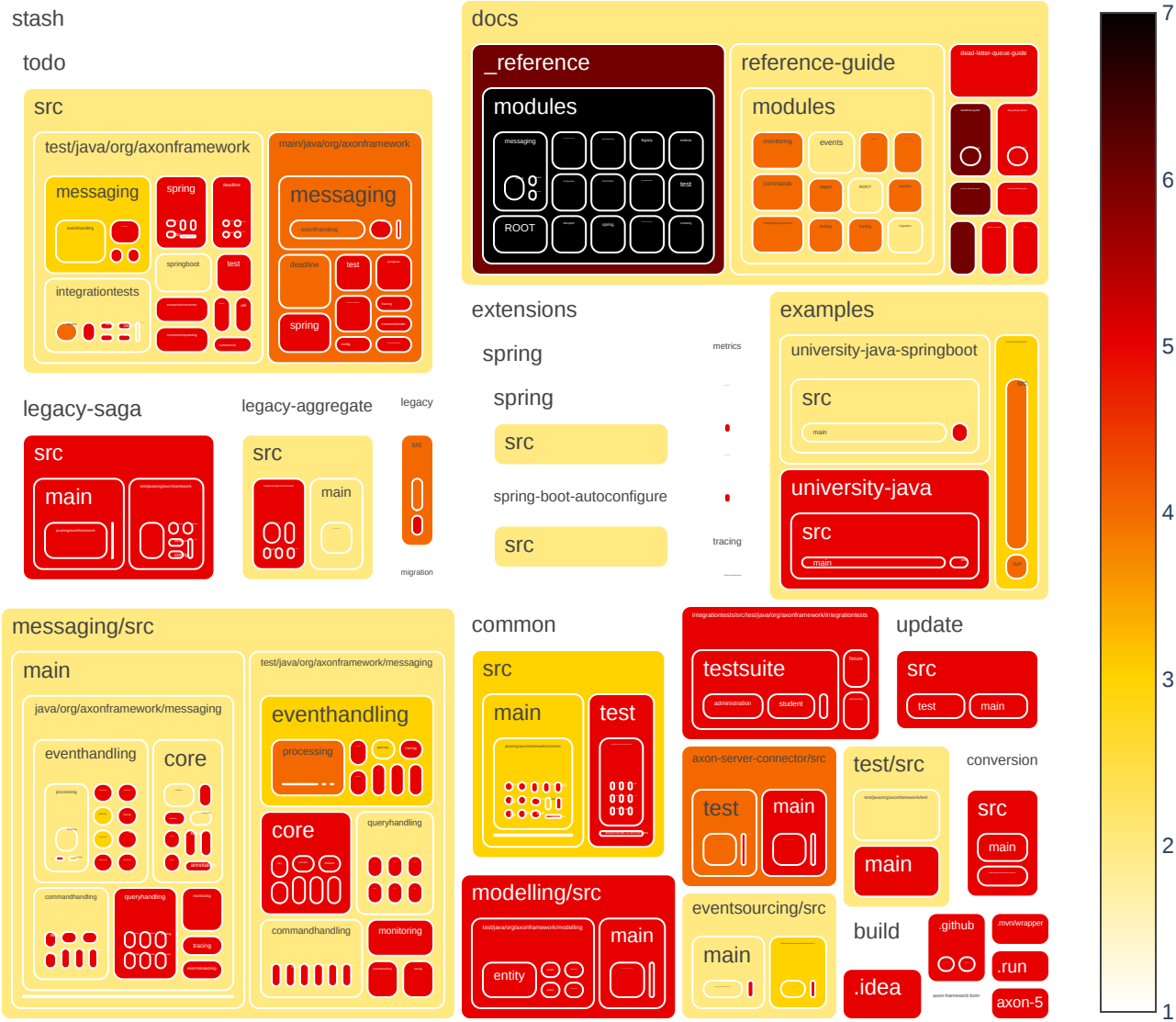
AxonFramework-5.0.2



Days since last commit per directory (ranked)

Rank of days since last commit

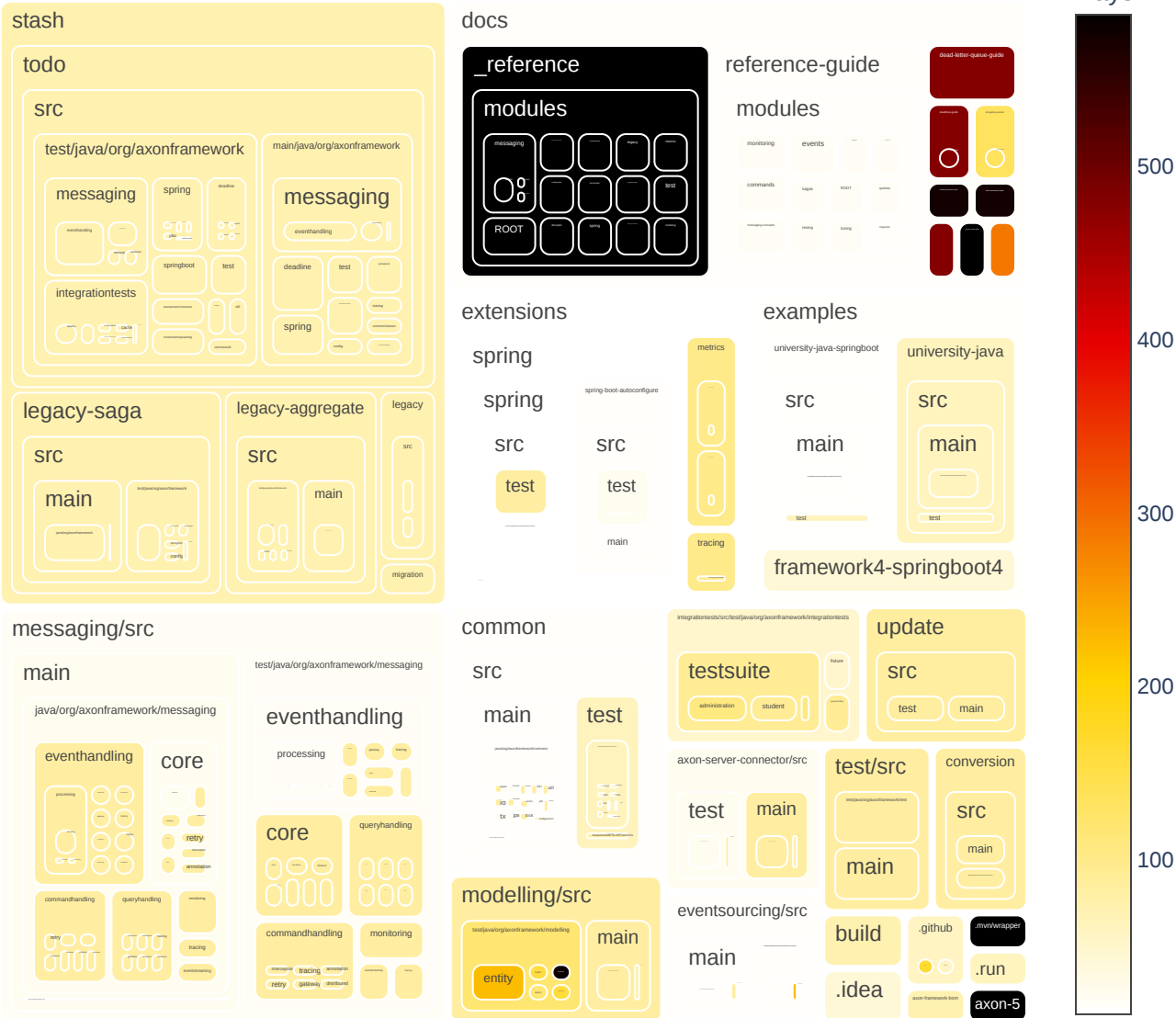
AxonFramework-5.0.2



Days since last file creation per directory

Days since last file creation

AxonFramework-5.0.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-5.0.2



Days since last file modification per directory

Days since last file modification

AxonFramework-5.0.2

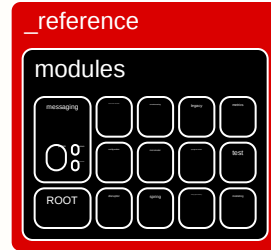
stash

todo

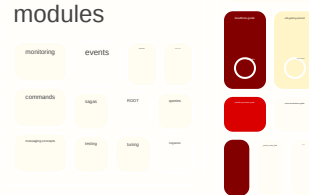
src



docs



reference-guide



Days

500

400

300

200

100

legacy-saga



legacy-aggregate



legacy

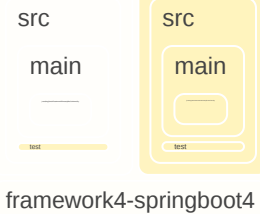
extensions

spring



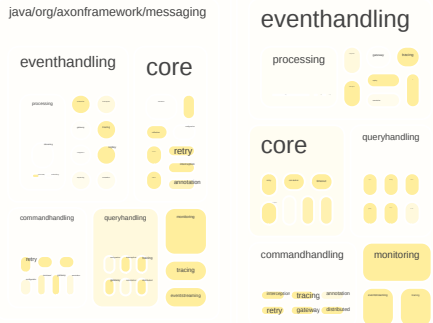
examples

spring-boot-autoconfigure



messaging/src

main



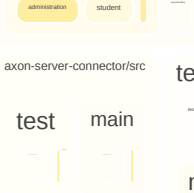
common

src

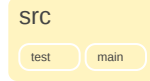


integrationtests/src/test/java/org/axonframework/integrationtests

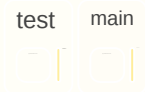
testsuite



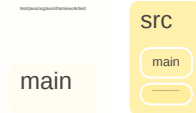
update



axon-server-connector/src



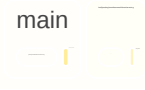
test/src



conversion



eventsourcing/src



build



modelling/src



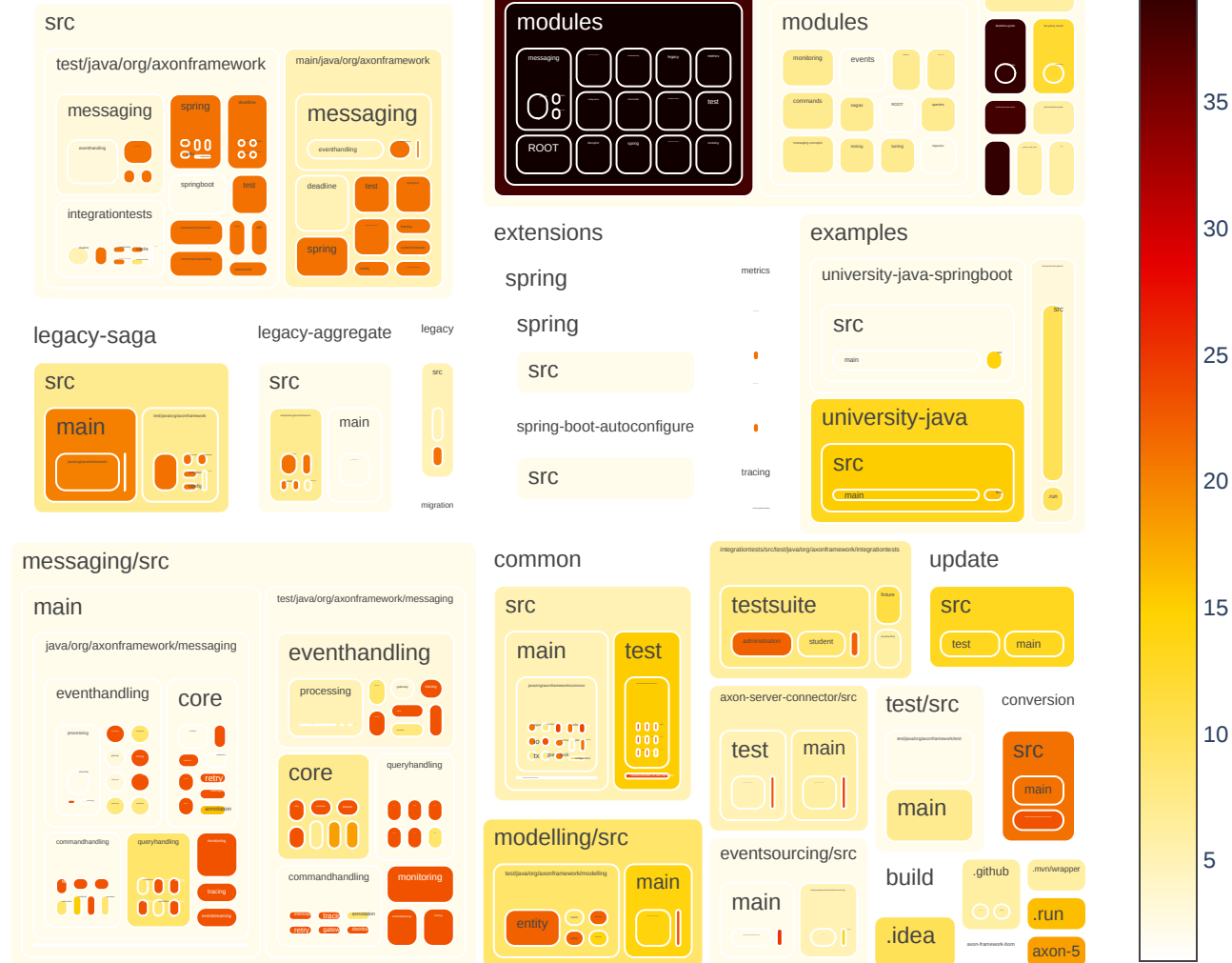
Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-5.0.2

stash

todo



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 1679

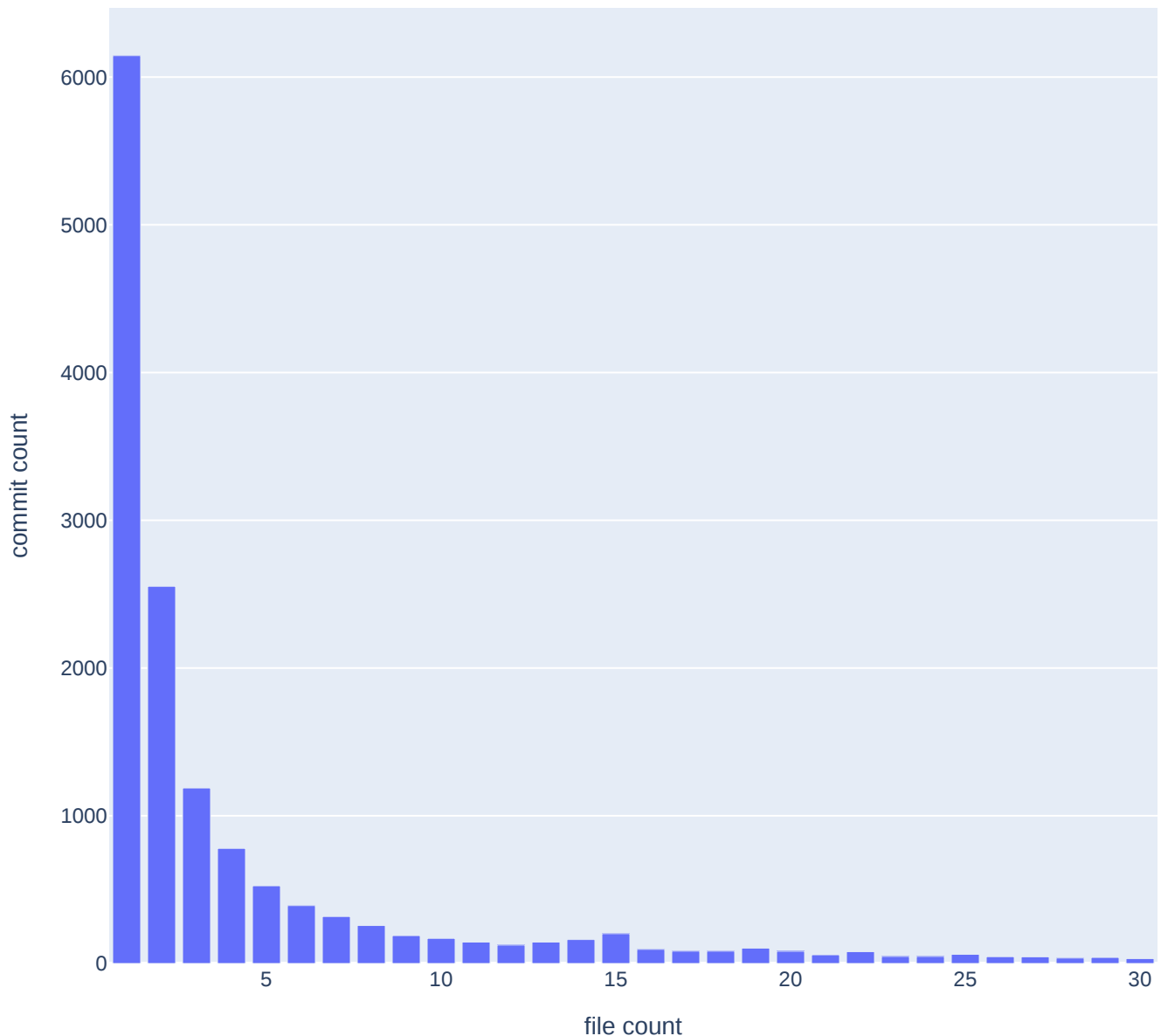
Max changed files with one commit = 4781

	filesPerCommit	commitCount
count	475.000000	475.000000
mean	572.324211	33.526316
std	826.005356	313.673903
min	1.000000	1.000000
25%	119.500000	1.000000
50%	260.000000	2.000000
75%	593.000000	5.000000
max	4781.000000	6145.000000

	filesPerCommit	commitCount
0	1	6145
1	2	2553
2	3	1188
3	4	779
4	5	526
5	6	392
6	7	318
7	8	257
8	9	189
9	10	170
10	11	145
11	12	125
12	13	145
13	14	162
14	15	202
15	16	96
16	17	84
17	18	84
18	19	104
19	20	83
20	21	59
21	22	80
22	23	48
23	24	48
24	25	62
25	26	46
26	27	45
27	28	37
28	29	41
29	30	33

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

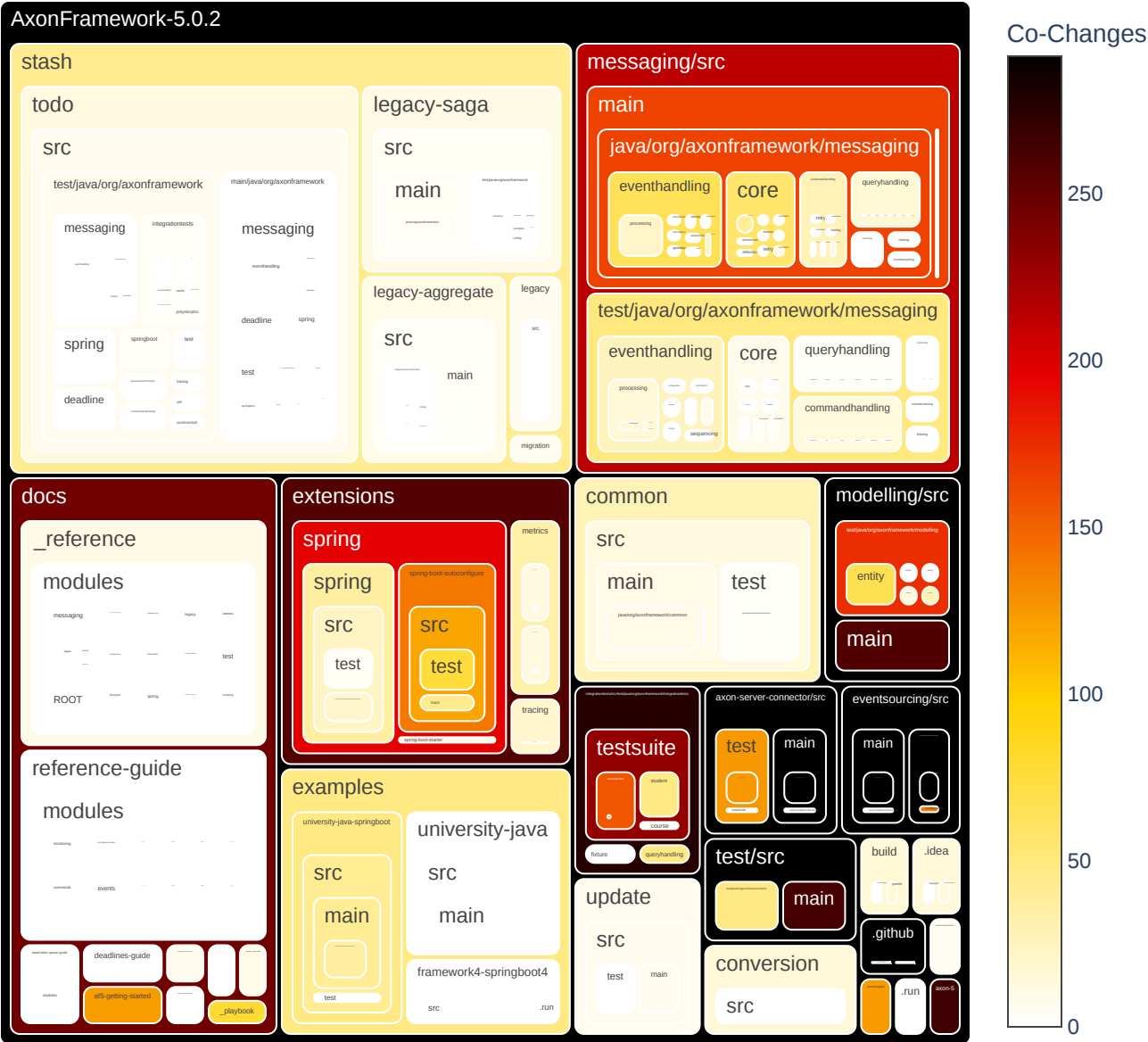
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

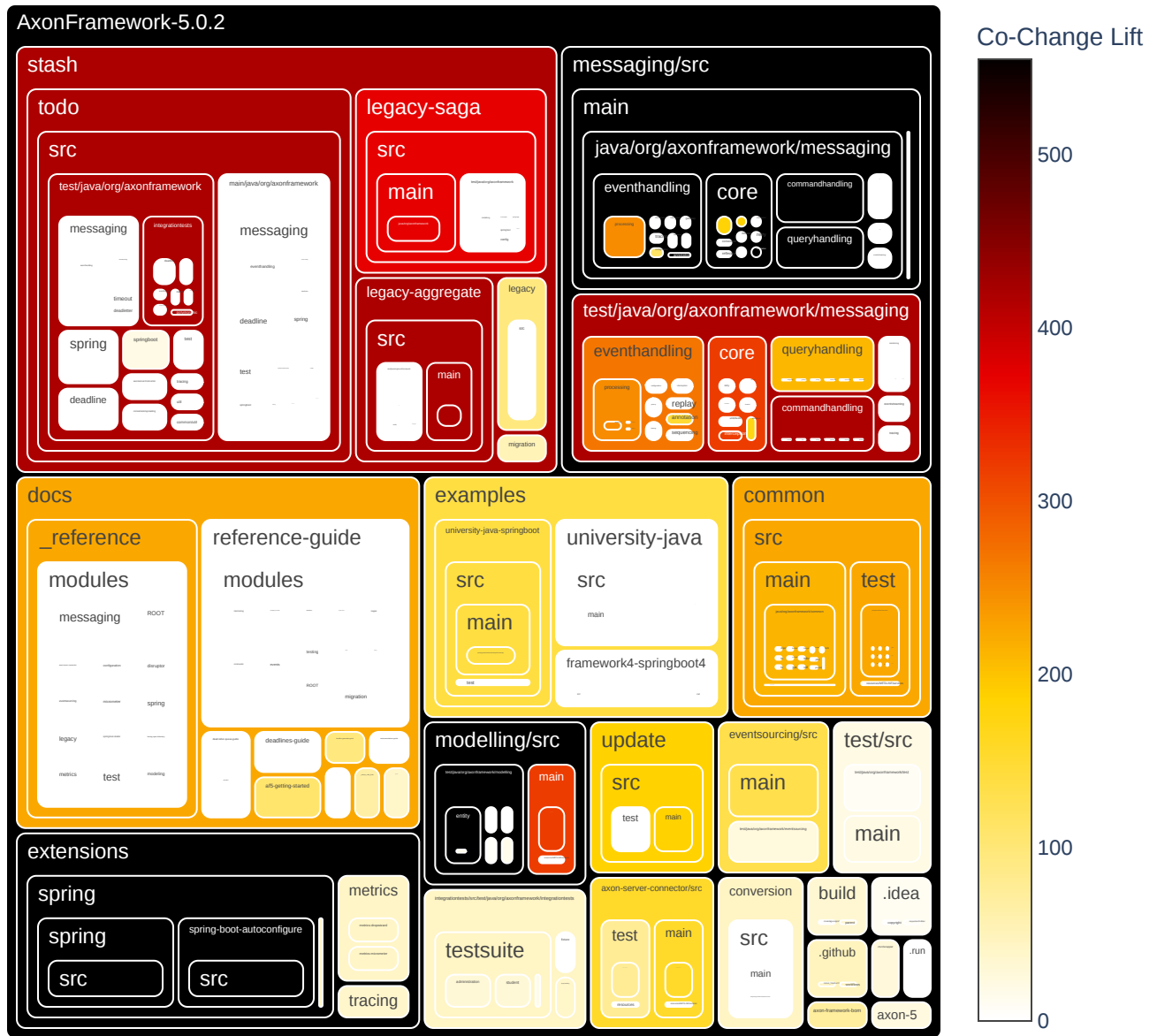
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

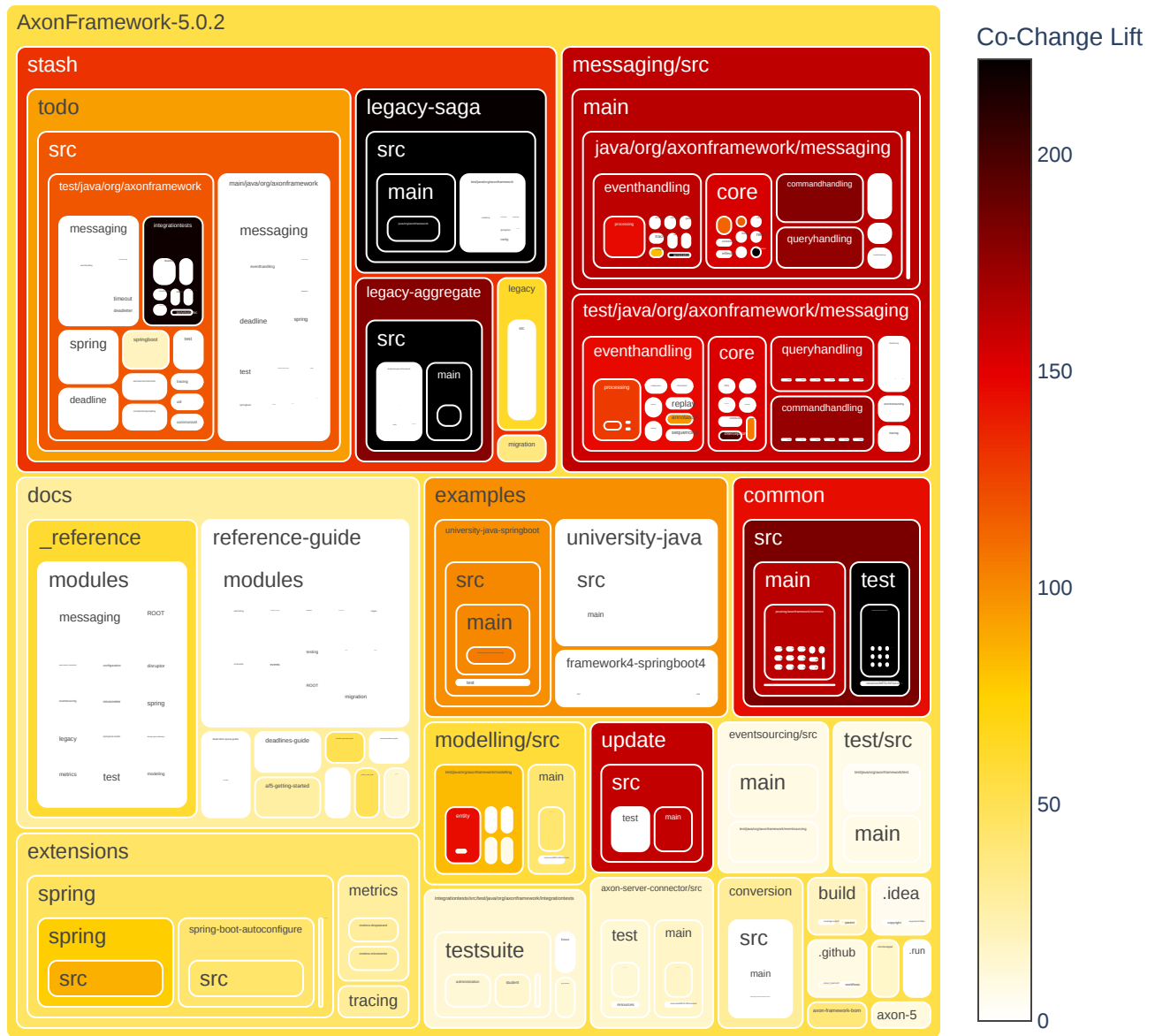
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

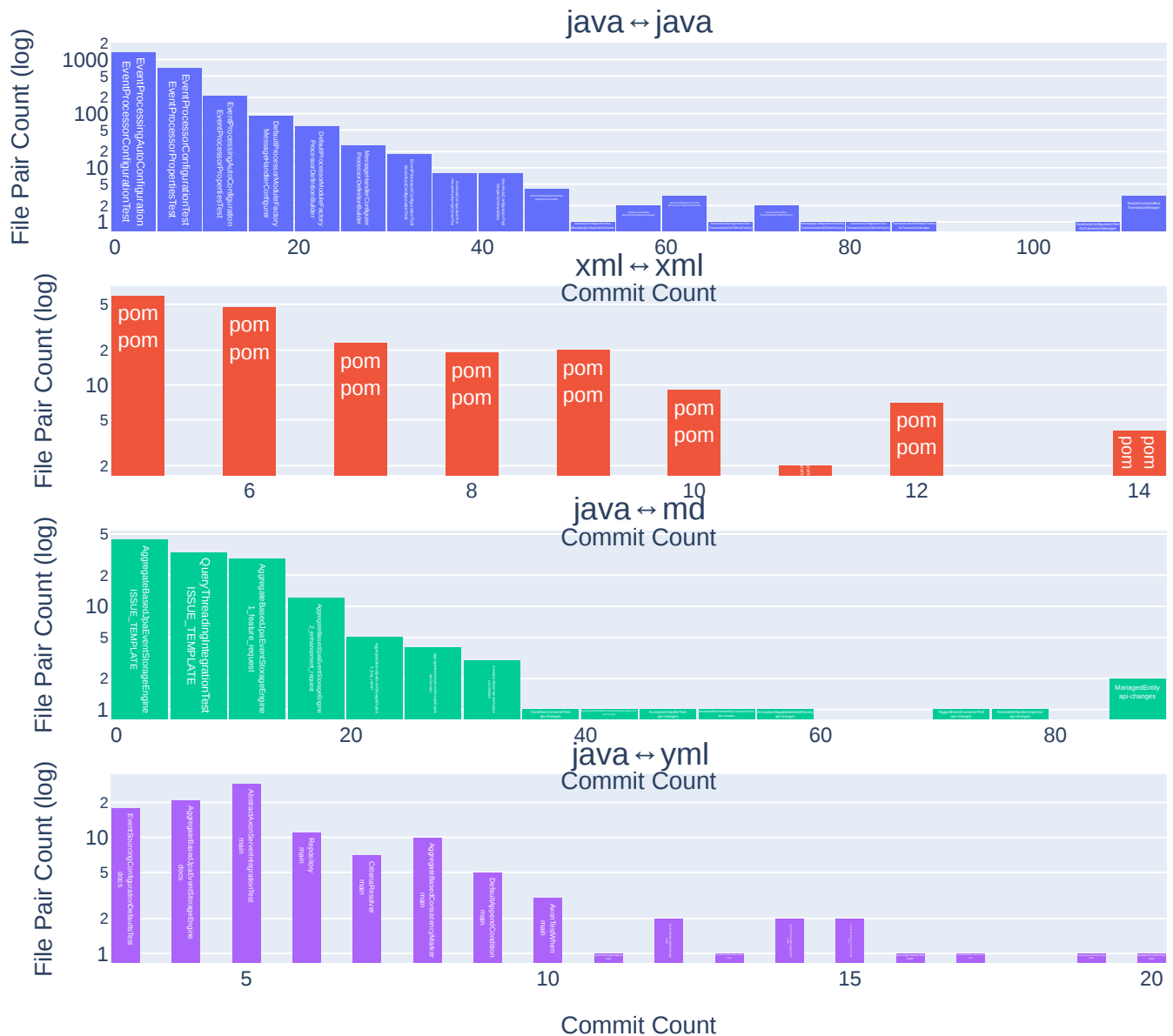


	fileExtensionPair	fileExtensionPairCount
0	java↔java	2521
1	xml↔xml	190
2	java↔md	139
3	java↔yaml	116

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair
0	java↔java	114	1	StorageEngineTestSuite↔AggregateBasedStorageEngineTestSuite
1	java↔java	112	2	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine connector
2	java↔java	111	3	AggregateBasedJpaEventStorageEngine↔AggregateBasedStorageEngineTestSuite event
3	java↔java	109	4	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedStorageEngineTestSuite connector
4	java↔java	85	5	StorageEngineTestSuite↔AggregateBasedJpaEventStorageEngine
5	java↔java	83	6	StorageEngineTestSuite↔AggregateBasedAxonServerEventStorageEngine event
6	java↔java	76	7	StorageEngineTestSuite↔DefaultEventStoreTransactionTest
7	java↔java	73	8	AggregateBasedStorageEngineTestSuite↔DefaultEventStoreTransactionTest
8	java↔java	72	9	StorageEngineTestSuite↔DefaultEventStoreTransaction
9	java↔java	65	10	DefaultEventStoreTransaction↔AggregateBasedStorageEngineTestSuite
10	xml↔xml	14	1	pom↔pom
11	xml↔xml	12	2	pom↔pom
12	xml↔xml	11	3	pom↔pom
13	xml↔xml	10	4	pom↔pom
14	xml↔xml	9	5	pom↔pom
15	xml↔xml	8	6	pom↔pom
16	xml↔xml	7	7	pom↔pom
17	xml↔xml	6	8	pom↔pom
18	xml↔xml	5	9	pom↔pom
19	java↔md	87	1	AggregateBasedStorageEngineTestSuite↔api-changes
20	java↔md	86	2	AggregateBasedAxonServerEventStorageEngine↔api-changes
21	java↔md	78	3	AggregateBasedJpaEventStorageEngine↔api-changes
22	java↔md	70	4	StorageEngineTestSuite↔api-changes
23	java↔md	55	5	SimpleEventStoreTest↔api-changes
24	java↔md	51	6	DefaultEventStoreTransactionTest↔api-changes
25	java↔md	46	7	QueryThreadingIntegrationTest↔api-changes
26	java↔md	41	8	DefaultEventStoreTransaction↔api-changes
27	java↔md	39	9	EventSourcingConfigurationDefaults↔api-changes
28	java↔md	34	10	AxonServerEventStorageEngine↔api-changes
29	java↔yaml	20	1	DefaultEventStoreTransaction↔pullrequest
30	java↔yaml	19	2	StorageEngineTestSuite↔pullrequest
31	java↔yaml	17	3	AggregateBasedStorageEngineTestSuite↔pullrequest
32	java↔yaml	16	4	AggregateBasedStorageEngineTestSuite↔main
33	java↔yaml	15	5	StorageEngineTestSuite↔main
34	java↔yaml	14	6	DefaultEventStoreTransaction↔main
35	java↔yaml	13	7	QueryThreadingIntegrationTest↔pullrequest
36	java↔yaml	12	8	QueryThreadingIntegrationTest↔main
37	java↔yaml	11	9	AppendCondition↔pullrequest
38	java↔yaml	10	10	NoAppendCondition↔main

Co-Changed Files by their commit count

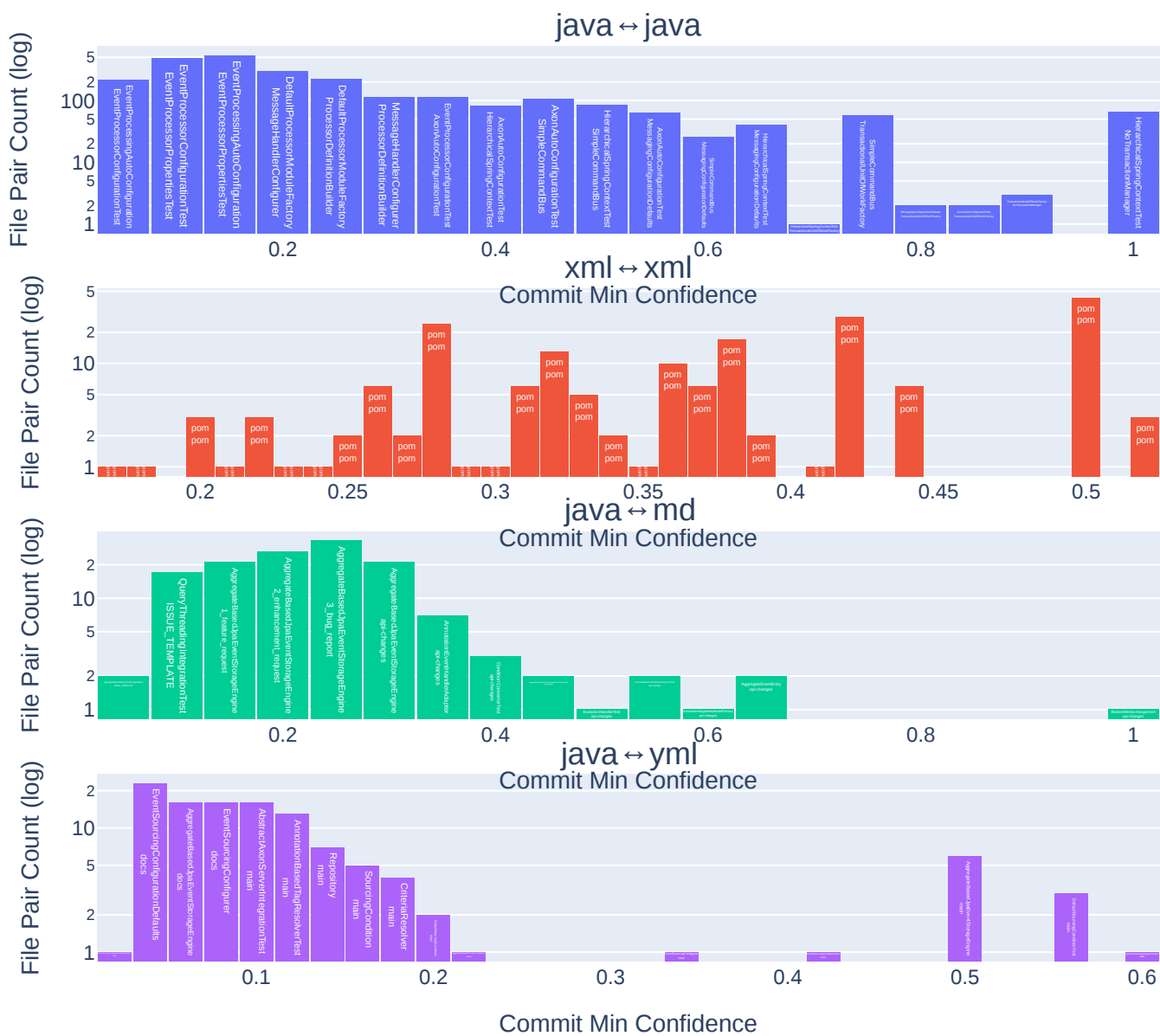


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	
0	java↔java	1.000000	1	DefaultProcessorModuleFactory↔MessageHandlerConfigurer extensio
1	java↔java	0.900000	2	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition
2	java↔java	0.875000	3	ChildEntityFieldDefinition↔FieldChildEntityFieldDefinition
3	java↔java	0.833333	4	AnnotatedEventHandlingComponent↔AnnotationEventHandlerAdapter messaging/src/mai
4	java↔java	0.800000	5	ConditionConverter↔SpringUtils
5	java↔java	0.750000	6	MessagingConfigurationDefaults↔CommandBusTestUtils
6	java↔java	0.714286	7	FutureUtils↔DefaultAxonApplication
7	java↔java	0.666667	8	DefaultAxonApplication↔DefaultAxonApplicationTest
8	java↔java	0.661538	9	DefaultSourcingConditionTest↔SourcingConditionTest
9	java↔java	0.648148	10	DefaultAppendConditionTest↔NoAppendConditionTest
10	xml↔xml	0.518519	1	pom↔pom
11	xml↔xml	0.500000	2	pom↔pom
12	xml↔xml	0.444444	3	pom↔pom
13	xml↔xml	0.416667	4	pom↔pom
14	xml↔xml	0.407407	5	pom↔pom
15	xml↔xml	0.392857	6	pom↔pom
16	xml↔xml	0.388889	7	pom↔pom
17	xml↔xml	0.384615	8	pom↔pom
18	xml↔xml	0.370370	9	pom↔pom
19	xml↔xml	0.363636	10	pom↔pom
20	java↔md	1.000000	1	AnnotationSagaMetaModelFactory↔api-changes
21	java↔md	0.666667	2	AnnotationEventHandlerAdapter↔api-changes
22	java↔md	0.600000	3	CommandConverter↔modularity
23	java↔md	0.571429	4	ExceptionHandlerTest↔api-changes
24	java↔md	0.500000	5	SimpleCommandHandlingComponentTest↔api-changes
25	java↔md	0.445596	6	AggregateBasedAxonServerEventStorageEngine↔api-changes
26	java↔md	0.441176	7	ChildAmbiguityException↔api-changes
27	java↔md	0.391304	8	Repository↔api-changes
28	java↔md	0.375000	9	FieldChildEntityFieldDefinition↔api-changes
29	java↔md	0.367925	10	AggregateBasedJpaEventStorageEngine↔api-changes
30	java↔yaml	0.600000	1	SpringUtils↔slack-release-notification
31	java↔yaml	0.555556	2	RenameCourseController↔application
32	java↔yaml	0.500000	3	UnsubscribeStudentToCourseController↔application exampl
33	java↔yaml	0.428571	4	EventProcessingAutoConfiguration↔test-docker-compose
34	java↔yaml	0.333333	5	UpdateCheckerProperties↔application
35	java↔yaml	0.222222	6	NoAppendCondition↔pullrequest
36	java↔yaml	0.203390	7	NoAppendConditionTest↔pullrequest
37	java↔yaml	0.193548	8	DefaultEventStoreTransaction↔add-to-project
38	java↔yaml	0.185185	9	DefaultAppendConditionTest↔pullrequest
39	java↔yaml	0.183333	10	AppendCondition↔pullrequest

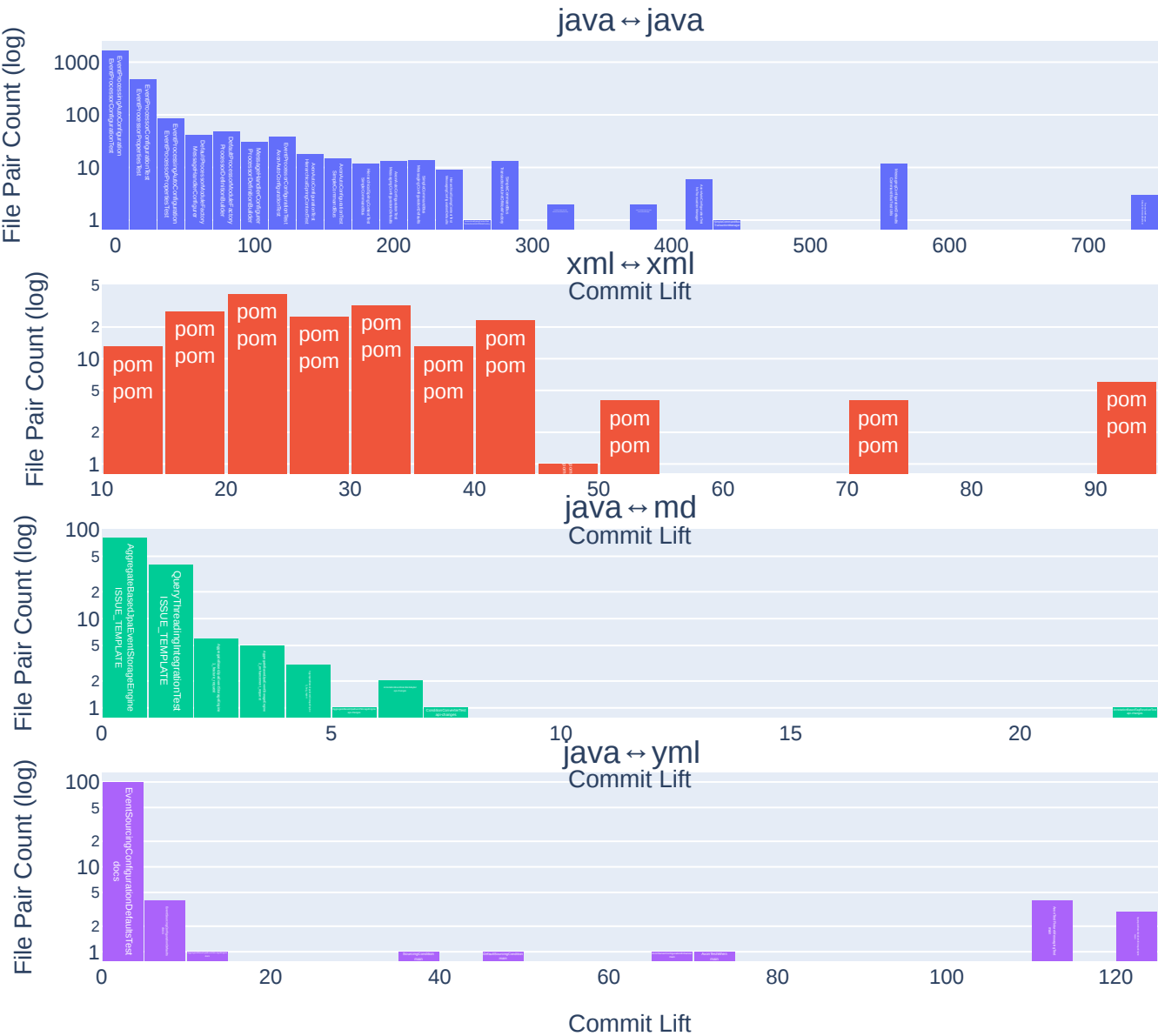
Co-Changed Files by their commit min confidence



Files changed together by commit lift

	fileExtensionPair	updateCommitLift	GroupRank	filePair
0	java↔java	740.000000	1	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest
1	java↔java	555.000000	2	Command↔Event
2	java↔java	555.000000	3	DefaultProcessorModuleFactory↔ProcessorDefinitionBuilder
3	java↔java	444.000000	4	MethodSequencingPolicyEventHandlerDefinition↔MethodSequencingPolicyEventHandlerDefinitionTest
4	java↔java	416.250000	5	CommandBusTestUtils↔AggregateTestFixture
5	java↔java	370.000000	6	HandlerComparator↔AnnotationSagaMetaModelFactory
6	java↔java	317.142857	7	AnnotationBasedEntityEvolvingComponent↔AnnotationSagaMetaModelFactory
7	java↔java	277.500000	8	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition
8	java↔java	277.500000	9	CommandBusTestUtils↔SimpleCommandBusTest
9	java↔java	266.400000	10	SimpleEventHandlingComponentSequencingPolicyTest↔SimpleEventHandlingComponentTest
10	xml↔xml	92.500000	1	pom↔pom
11	xml↔xml	71.153846	2	pom↔pom
12	xml↔xml	51.388889	3	pom↔pom
13	xml↔xml	47.435897	4	pom↔pom
14	xml↔xml	42.692308	5	pom↔pom
15	xml↔xml	42.633745	6	pom↔pom
16	xml↔xml	41.111111	7	pom↔pom
17	xml↔xml	41.111111	8	pom↔pom
18	xml↔xml	37.000000	9	pom↔pom
19	xml↔xml	36.543210	10	pom↔pom
20	java↔md	22.965517	1	CommandConverter↔modularity
21	java↔md	7.235984	2	NoAppendConditionTest↔design-principles
22	java↔md	6.776557	3	NoAppendCondition↔design-principles
23	java↔md	6.324786	4	DefaultAppendConditionTest↔design-principles
24	java↔md	5.692308	5	AppendCondition↔design-principles
25	java↔md	4.851870	6	QueryThreadingIntegrationTest↔README
26	java↔md	4.833091	7	AxonServerMessageStream↔design-principles
27	java↔md	4.065934	8	AccessSerializingRepository↔design-principles
28	java↔md	3.940828	9	DefaultSourcingConditionTest↔design-principles
29	java↔md	3.881119	10	SourcingConditionTest↔design-principles
30	java↔yaml	123.333333	1	RenameCourseController↔application
31	java↔yaml	111.000000	2	UnsubscribeStudentToCourseController↔application
32	java↔yaml	74.000000	3	UpdateCheckerProperties↔application
33	java↔yaml	65.294118	4	UpdateCheckerAutoConfiguration↔application
34	java↔yaml	46.250000	5	EventProcessingAutoConfiguration↔application
35	java↔yaml	39.642857	6	EventProcessingAutoConfiguration↔test-docker-compose
36	java↔yaml	11.482759	7	SpringUtils↔slack-release-notification
37	java↔yaml	6.068890	8	NoAppendConditionTest↔add-to-project
38	java↔yaml	5.967742	9	AppendCondition↔add-to-project
39	java↔yaml	5.683564	10	NoAppendCondition↔add-to-project

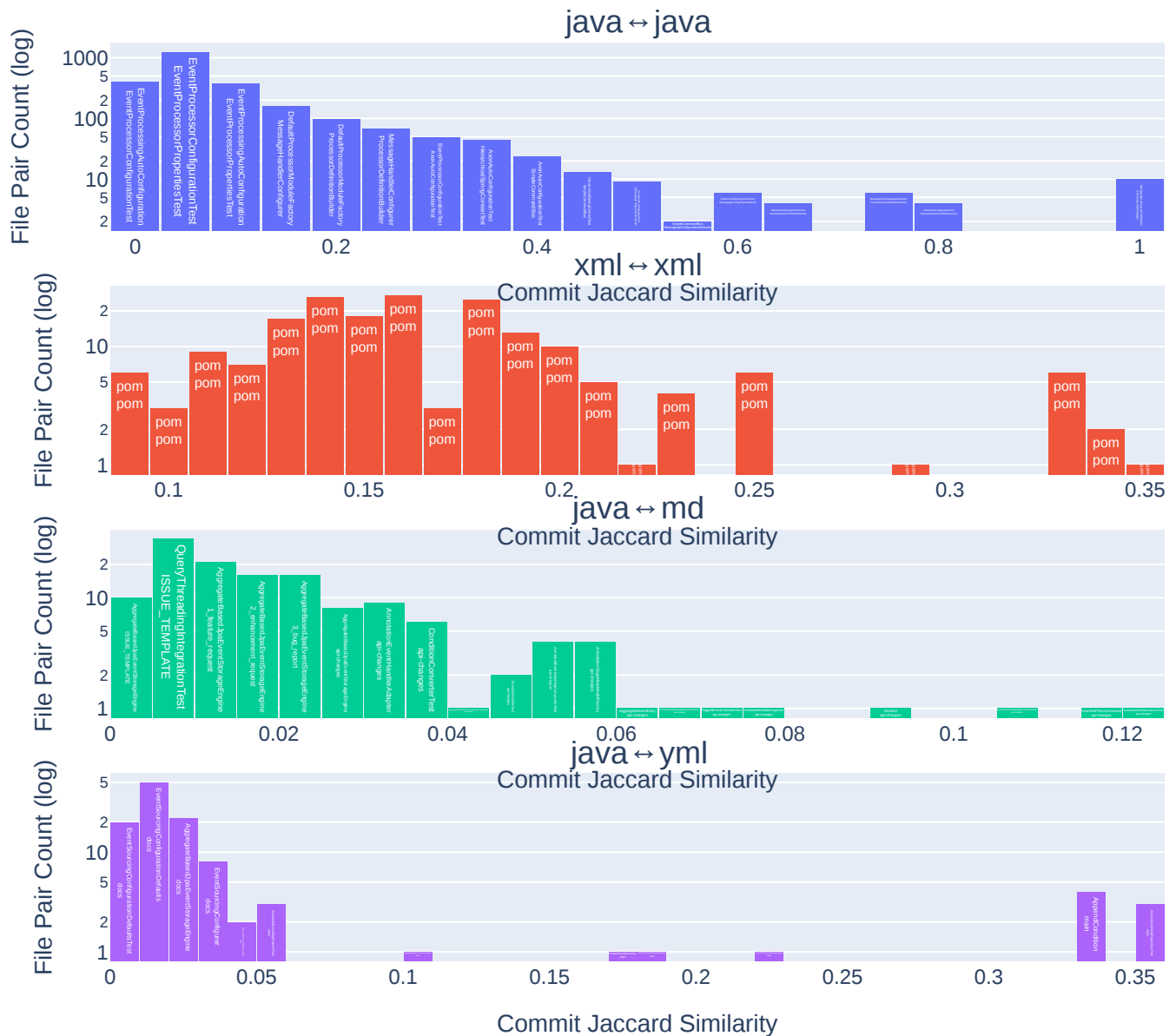
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	
0	java↔java	1.000000	1 Com
1	java↔java	0.818182	2 ChildEntityFieldDefinition↔GetterSetterChildEntityFi
2	java↔java	0.800000	3 MethodSequencingPolicyEventHandlerDefinition↔MethodSequencingPolicyEventHandlerC
3	java↔java	0.750000	4 DefaultProcessorModuleFactory↔ProcessorDefi
4	java↔java	0.666667	5 HandlerComparator↔AnnotationSagaMeta
5	java↔java	0.636364	6 ChildEntityFieldDefinition↔FieldChildEntityFi
6	java↔java	0.600000	7 CommandBusTestUtils↔Aggrega
7	java↔java	0.571429	8 AnnotationBasedEntityEvolvingComponent↔AnnotationSagaMeta
8	java↔java	0.500000	9 MergeTa
9	java↔java	0.488636	10 DefaultSourcingConditionTest↔SourcingC
10	xml↔xml	0.350000	1
11	xml↔xml	0.341463	2
12	xml↔xml	0.333333	3
13	xml↔xml	0.285714	4
14	xml↔xml	0.250000	5
15	xml↔xml	0.227273	6
16	xml↔xml	0.219512	7
17	xml↔xml	0.214286	8
18	xml↔xml	0.209302	9
19	xml↔xml	0.200000	10
20	java↔md	0.122682	1 AggregateBasedAxonServerEventStorageEngine↔
21	java↔md	0.116000	2 AggregateBasedStorageEngineTestSuite↔
22	java↔md	0.107143	3 AggregateBasedJpaEventStorageEngine↔
23	java↔md	0.091864	4 StorageEngineTestSuite↔
24	java↔md	0.077247	5 SimpleEventStoreTest↔
25	java↔md	0.071529	6 DefaultEventStoreTransactionTest↔
26	java↔md	0.066667	7 QueryThreadingIntegrationTest↔
27	java↔md	0.062500	8 NoAppendConditionTest↔desi
28	java↔md	0.059524	9 NoAppendCondition↔desi
29	java↔md	0.057665	10 DefaultEventStoreTransaction↔
30	java↔yaml	0.357143	1 RenameCourseController
31	java↔yaml	0.333333	2 UnsubscribeStudentToCourseController
32	java↔yaml	0.227273	3 UpdateCheckerAutoConfiguration
33	java↔yaml	0.187500	4 UpdateCheckerProperties
34	java↔yaml	0.172414	5 EventProcessingAutoConfiguration
35	java↔yaml	0.107143	6 EventProcessingAutoConfiguration↔test-doc
36	java↔yaml	0.058824	7 NoAppendConditionTest↔ai
37	java↔yaml	0.058140	8 AppendCondition↔ai
38	java↔yaml	0.056180	9 NoAppendCondition↔ai
39	java↔yaml	0.049383	10 DefaultAppendConditionTest↔ai

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountEx
0	java↔java	Command↔Event	60	
1	java↔java	Command↔Query	60	
2	java↔java	Event↔Query	60	
3	java↔java	Event↔QueryResponse	60	
4	java↔java	Query↔QueryResponse	60	
5	java↔java	Command↔QueryResponse	60	
6	java↔java	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	60	
7	java↔java	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	60	
8	java↔java	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	60	
9	java↔java	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition	62	
10	java↔md	AggregateBasedAxonServerEventStorageEngine↔api-changes	28	
11	java↔md	AggregateBasedJpaEventStorageEngine↔api-changes	39	
12	java↔md	AggregateBasedStorageEngineTestSuite↔api-changes	40	
13	java↔md	CommandConverter↔modularity	53	
14	java↔md	SimpleEventStoreTest↔api-changes	57	
15	java↔md	QueryThreadingIntegrationTest↔api-changes	59	
16	java↔md	QueryThreadingIntegrationTest↔README	63	
17	java↔md	AxonServerEventStorageEngine↔api-changes	67	
18	java↔md	DefaultEventStoreTransactionTest↔api-changes	70	
19	java↔md	StorageEngineTestSuite↔api-changes	71	
20	java↔yaml	RenameCourseController↔application	19	
21	java↔yaml	CreateCourseController↔application	19	
22	java↔yaml	ChangeCourseCapacityController↔application	19	
23	java↔yaml	UnsubscribeStudentToCourseController↔application	22	
24	java↔yaml	EnrollStudentInFacultyController↔application	22	
25	java↔yaml	CourseStatsController↔application	22	
26	java↔yaml	SubscribeStudentToCourseController↔application	22	
27	java↔yaml	UpdateCheckerAutoConfiguration↔application	25	
28	java↔yaml	EventProcessingAutoConfiguration↔application	28	
29	java↔yaml	UpdateCheckerProperties↔application	29	
30	xml↔xml	pom↔pom	9	
31	xml↔xml	pom↔pom	11	
32	xml↔xml	pom↔pom	11	
33	xml↔xml	pom↔pom	14	
34	xml↔xml	pom↔pom	14	
35	xml↔xml	pom↔pom	14	
36	xml↔xml	pom↔pom	14	
37	xml↔xml	pom↔pom	14	
38	xml↔xml	pom↔pom	14	
39	xml↔xml	pom↔pom	19	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitCount
0	pom↔pom	9	14	
1	AbstractConsistencyMarker↔ConsistencyMarker	343	10	
2	ConsistencyMarker↔ConsistencyMarkers	289	17	
3	AggregateBasedAxonServerEventStorageEngine↔AggregateBasedJpaEventStorageEngine	668	112	
4	EventSourcingConfigurationDefaults↔EventSourcingConfigurationDefaultsTest	731	53	
5	StorageEngineTestSuite↔AggregateBasedStorageEngineTestSuite	856	114	
6	DefaultAppendConditionTest↔NoAppendConditionTest	222	35	
7	DefaultSourcingConditionTest↔SourcingConditionTest	232	43	
8	Command↔Event	60	4	
9	Command↔Query	60	4	
10	Event↔Query	60	4	
11	Event↔QueryResponse	60	4	
12	Query↔QueryResponse	60	4	
13	Command↔QueryResponse	60	4	
14	FieldChildEntityFieldDefinition↔GetterEvolverChildEntityFieldDefinition	62	8	
15	ChildEntityFieldDefinition↔GetterSetterChildEntityFieldDefinition	73	9	
16	FieldChildEntityFieldDefinitionTest↔GetterEvolverChildEntityFieldDefinitionTest	60	3	
17	GetterEvolverChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	60	3	
18	FieldChildEntityFieldDefinitionTest↔GetterSetterChildEntityFieldDefinitionTest	60	3	

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCount	updateCommitCount
0	java↔java	AggregateBasedJpaEventStorageEngine↔EventSourcingConfigurationDefaultsTest	3230	4	
1	java↔java	AxonServerCommandBusConnector↔AggregateBasedStorageEngineTestSuite	3257	3	
2	java↔java	EventSourcingConfigurationDefaults↔QueryThreadingIntegrationTest	3209	4	
3	java↔java	QueryThreadingIntegrationTest↔AnnotationBasedEntityEvolvingComponentTest	3183	3	

WordCloud of git authors

	word	frequency
0	Steven van Beelen	6273
1	Allard Buijze	2928
2	smcvb	822
3	Mitchell Herrijgers	741
4	Mateusz Nowak	632
5	Milan Savic	449
6	Gerard Klijs	246
7	Marc Gathier	231
8	Simon Zambrovski	226
9	sara pellegrini	163

Wordcloud of git authors

